

# Day 32 - IPV6 (pt 2)

Sunday, September 6, 2026 10:46 AM

## IPv6 Information

### EUI-64

'EUI-64 / modified EUI (Extended Unique Identifier)' is a method of converting MAC addresses (48 bits) into a 64-bit interface identifier. The resulting interface identifier can then become the 'host portion' of a /64 IPv6 address.

- Steps to convert a MAC address into EUI-64
  - Divide the MAC address in half
    - 1234 5678 90AB --> 1234 56 | 78 90AB
  - Insert FFFE in the middle.
    - 1234 56FF FE78 90AB
  - Invert the 7th bit
    - 1234 56FF FE78 90AB --> 1034 56FF FE78 90AB

Why is the 7th bit inverted?

- MAC addresses can be split into two types:
  - UAA (Universally Administered Address)
    - Globally unique address that is assigned to the device by the manufacturer
  - LAA (Locally Administered Address)
    - Manually assigned by an admin (through the mac-address command on the interface) or protocol. Doesn't have to be globally unique.
    - You can identify UAA and LAA addresses through the 7th bit of the address, called a U/L bit (Universal/Local bit)
      - U/L bit = 0 --> Universal address
      - U/L bit = 1 --> Local address
    - In the context of IPv6 addresses/EUI-64, the meaning of the U/L bit is reversed.
      - U/L bit = 0 --> The MAC address the EUI-64 interface ID was made from was an LAA
      - U/L bit = 1 --> The MAC address the EUI-64 interface ID was made from was an UAA

## Types of Addresses

### 1. Global Unicast Addresses

Global Unicast addresses are public addresses which can be used over the internet. Must be globally unique.

- Ex. Address:
  - 2001 : 0DB8 : 8B00 : 0001 : 0000 : 0000 : 0000 : 0001 /64
- Range:
  - 2000::/3 (Later changed to be all addresses which aren't reserved for other purposes)
- The Global Unicast address consists of 3 different sections:
  - Global Routing Prefix (48 bits)
    - A prefix that's assigned by the ISP
    - Its location in the previous example: '2001 : 0DB8 : 8B00'
  - Subnet Identifier (16 bits)
    - Used to make various subnets
    - Its location in the previous example: '0001'
  - Interface Identifier (64 bits)
    - Represents the host portion of the address
    - Its location in the previous example: '0000 : 0000 : 0000 : 0001'

|                                      |
|--------------------------------------|
|                                      |
| Global Unicast address photo summary |

### 2. Unique Local Addresses

Unique Local addresses are private addresses which cannot be used or routed over the internet. They are used in internal networks and don't need to be globally unique. (It's recommended they're unique in the local network)

- Ex. Address:
  - FD45 : 93AC : 8A8F : 0001 : 0000 : 0000 : 0000 : 0001 /64
- Range:
  - FC00::/7 (Later update requires the 8th bit to be set to 1, so the first two hex digits must be FD)
- The Unique Local address consists of 4 different sections:
  - Hexadecimal FD (8 bits)
    - Indicates a unique local address
    - Its location in the previous example: 'FD'
  - Global ID (40 bits)
    - A prefix that should be randomly generated
    - Its location in the previous example: '45 : 93AC : 8A8F'
  - Subnet Identifier (16 bits)
    - Used to make various subnets
    - Its location in the previous example: '0001'

- Its location in the previous example: '0001'

#### 4. Interface Identifier (64 bits)

- Represents the host portion of the address
- Its location in the previous example: '0000 : 0000 : 0000 : 0001'

|                                    |
|------------------------------------|
|                                    |
| Unique Local address photo summary |

### 3. Link Local Addresses

Link-local are IPv6 addresses are automatically generated on all IPv6-enabled interfaces, even ones enabled through the [ipv6 enable command](#). Link-local means that these addresses are used for communications within a [single link \(subnet\)](#). Routers will not forward packets with a link-local destination IPv6 address.

- **Ex. Address:**
  - FE80 :: EF8 : 22FF : FE36 : 8500
- **Range:**
  - FE80::/10
  - (Standard states that the 54 bits after FE80::/10 should all be 0, so link-local addresses always begin with FE8.
- The **Link-local** address consists of 3 different sections:
  1. **FE80 (16 bits)**
    - Indicates a **link-local** address.
    - Its location in the previous example: 'FE80'
  2. **:: (48 bits)**
    - The section after the FE80.
    - Set to all **zeroes** and represented by a double colon ( :: )
  3. **EUI-64 Interface Identifier (64 bits)**
    - Represents the host portion of the address, and is randomly generated using **EUI-64**
    - Its location in the previous example: 'EF8 : 22FF : FE36 : 8500'

#### Uses for Link Local Addresses

A few common uses for link-local addresses include:

1. Routing protocol peerings (**OSPFv3** uses **link-local** addresses for neighbor adjacencies)
2. Next-hop addresses for static routes
3. **Neighbor Discovery Protocol (NDP)**, IPv6's replacement for **ARP** uses **link-local** addresses to function [ipv6 enable and Link-Local addresses](#)

When the [ipv6 enable](#) command is used on an interface, it will automatically generate only a link-local address on the interface.

To assign a global unicast or unique local address to the enabled interface, you'd have to use the [ipv6 address PREFIX/PREFIX-LENGTH](#) command.

### 4. Multicast Addresses

Multicast addresses are used to send one-to-*many* messages.

- **Range:**
  - FF00::/8

#### IPv6 & IPv4 Multicast Addresses Table

| Purpose                                 | IPv6 Address    | IPv4 Address       |
|-----------------------------------------|-----------------|--------------------|
| All <b>nodes/hosts</b> (like broadcast) | FF02:: <b>1</b> | 224.0.0. <b>1</b>  |
| All <b>routers</b>                      | FF02:: <b>2</b> | 224.0.0. <b>2</b>  |
| All <b>OSPF</b> routers                 | FF02:: <b>5</b> | 224.0.0. <b>5</b>  |
| All <b>OSPF DR/BDR</b> routers          | FF02:: <b>6</b> | 224.0.0. <b>6</b>  |
| All <b>RIP</b> routers                  | FF02:: <b>9</b> | 224.0.0. <b>9</b>  |
| All <b>EIGRP</b> routers                | FF02: <b>A</b>  | 224.0.0. <b>10</b> |

| Purpose                                       | IPv6 Address   | IPv4 Address            |
|-----------------------------------------------|----------------|-------------------------|
| All nodes/hosts<br>(functions like broadcast) | FF02: <b>1</b> | 224 . 0 . 0 . <b>1</b>  |
| All routers                                   | FF02: <b>2</b> | 224 . 0 . 0 . <b>2</b>  |
| All OSPF routers                              | FF02: <b>5</b> | 224 . 0 . 0 . <b>5</b>  |
| All OSPF DRs/BDRs                             | FF02: <b>6</b> | 224 . 0 . 0 . <b>6</b>  |
| All RIP routers                               | FF02: <b>9</b> | 224 . 0 . 0 . <b>9</b>  |
| All EIGRP routers                             | FF02: <b>A</b> | 224 . 0 . 0 . <b>10</b> |

Small review on **-cast** addresses

- **Unicast** addresses are one-to-*one*
  - One source to *one* destination.
- **Broadcast** addresses are one-to-*all*
  - One source to *all* destinations (within the subnet).
- **Multicast** addresses are one-to-*many*

